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MOTHER–INFANT SEPARATION IN THE JAVA MONKEY (*MACACA IRUS*)[1]

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Twelve 7–8-mo.-old *Macaca irus* monkeys were separated from mothers and reunited. Infant playing and related behaviors decreased during the 3-wk. separation period. Presence of a conspecific unfamiliar adult female mother-substitute did not alleviate debilitating effects of separation. Effects were not attributable to traumatic process of forceful removal of mothers since a control group separated and immediately returned was minimally affected. During reunion, infants responded differentially to their own mothers regardless of whether they were returned to original or spatially reoriented home cages. Most behaviors returned to preseparation levels, but the mother–infant relationship was intensified in 3-wk.-separated animals. Species differences in reaction to maternal separation are discussed.

Using an apparatus in which mother–infant pairs of rhesus monkeys (*Macaca mulatta*) were housed on opposite ends of a play area to which only infants were allowed access, Seay, Hansen, and Harlow (1962) and Seay and Harlow (1965) found that maternal separation is extremely stressful to 6–7-mo.-old infants. When their mothers were not available, there was much crying (cooing vocalization) and a reduction in many complex social behaviors such as infant playing. Mothers reacted violently to being separated from their infants and attempted to maintain physical contact with them. At reunion, infants ran immediately to their mothers. There was an initial increase in infant–mother and mother–infant positive behavior, but the monkeys' behavior quickly returned to preseparation levels.

Basically similar results were obtained with infants separated at comparable ages by Spencer-Booth and Hinde (1967), also using the rhesus macaque, and by Kaufman and Rosenblum (1967) in the pigtail (*Macaca nemestrina*), but the increase in infant–mother and mother–infant behavior in reunion was not transient. Some of the infants in these investigations showed signs of prolonged disturbance even after several months following the mother's return. However, the monkeys in these studies were maintained in a complex group situation, and removal of mothers did not involve separation from other adults. It may be that the sheer number of adult animals in the group situation tends to exacerbate the effects of maternal separation in species in which infant interaction with adults other than the mother is limited. Rosenblum and Kaufman (1968) found that 2–10-mo.-old bonnet macaque (*Macaca radiata*) infants living in a group situation do not show the severe disturbance following maternal separation that seems to characterize the reactions of rhesus and pigtail monkeys. The investigators attribute the minimal reaction to separation to the permissive behavior of bonnet mothers who frequently allow other adults to explore and manipulate newborn infants almost immediately after birth. As a result, these infants become autonomous early and establish social relationships with adults other than their natural mother.

The present study investigated the generality of the effects of maternal separation in a different species of macaque, the Java monkey (*Macaca irus*). Maternal behavior and infant social development are quite similar in Java and rhesus monkeys (Seay, 1967) and it was expected that the reactions to maternal separation would also be quite similar.

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Contrast groups were included in the present study to evaluate the effects of the separation process itself and the effects of providing an unfamiliar conspecific mother substitute during separation. Following separation, the infants' ability to recognize and differentially respond to their own mothers was tested by changing the location of the cage occupied by their mothers.

METHOD

Subjects

Twelve mother-infant pairs of *M. irus* monkeys were divided into six groups of two mother-infant pairs each according to time of birth of infants. At the time of separation, infant ages ranged 208–255 days. The mean within-group age difference was 25 days with the maximum for any group being 38. Of the six groups employed, two contained a female–female infant pair (Groups A_1 and SM_1), two a male–male infant pair (Groups SM_2 and C_2), and two a male–female infant pair (Groups A_2 and C_1). All infants were laboratory born and all mothers were capative feral animals; previous maternal histories were unknown.

Apparatus

The apparatus is shown in Figure 1 and was constructed of galvanized, flattened, expanded metal on an angle-iron frame. It consisted of two 36 × 27 × 42 in. home cages attached on opposite ends of a 60 × 27 × 42 in. play area. The play area was further subdivided by a removable panel, and access to it was provided for infants only by 3.5-sq.-in. openings. Each home cage was equipped with a feeder, water bottle, wall-size front door, and transport door. The apparatus stood on legs 30 in. above the floor.

Procedure

Data collection was divided into three 3-wk. periods: preseparation, separation, and reunion. The preseparation period was the same for all groups and consisted for base-line observation of mothers and infants. During the 3-wk. separation period, infants in two groups were provided with adult mother substitutes (SM groups) while infants in another two groups (A groups) were not. Infants in the remaining two groups (C groups) were separated and immediately reunited with mothers. During reunion, mothers in half of the groups were returned to their original home cages (Groups A_1, SM_1, and C_1), while mothers in the remaining half were returned to identical but spatially reoriented home cages (Groups A_2, SM_2, and C_2).

The separation procedure involved the simultaneous placement of metal plates by two investigators between each home cage and the play area in such a way as to prevent infants from running to their mothers. A small transport cage was then placed on the edge of each home cage, and guillotine doors were raised to allow mothers to enter. The metal plates blocking the infants' access to home cages were then removed. The introduction of substitute mothers and the return of real mothers to home cages were also done by means of transport cages. Mothers were simultaneously released by two investigators while their infants were in the play area and the panel removed.

The panel subdividing the play area was removed for approximately ½ hr. per day, 4 days per week, during which time data collection was undertaken. Data collection consisted of eight 15-min. sessions per week on each mother-infant pair throughout the 9-wk. period. A modification of the symbol checklist developed by Hansen (1966) at the University of Wisconsin Primate Laboratory was used for this purpose. Mother and infant behavior was scored simultaneously as it has previously been shown that one observer is able to score two animals reliably and efficiently at the same time (Thorne, Schlottmann, & Seay, 1969). A tape-recorded voice counted from 0 to 60 with 15-sec. intervals between successive numbers to indicate when entries were to be made on the next row. The maximum score for each behavior occurring in a 15-sec. interval was 1. Therefore, the maximum score for a behavior category in the 15-min. session was 60. In recording data, observers were seated on stools approximately 4 ft. from the test apparatus and in full view of the subjects.

Description of Behavior

The following is a list of behaviors scored and their definitions. Two observers participated in data collection and estimates of interobserver reliability obtained by means of Pearson product-moment correlation coefficients are listed after

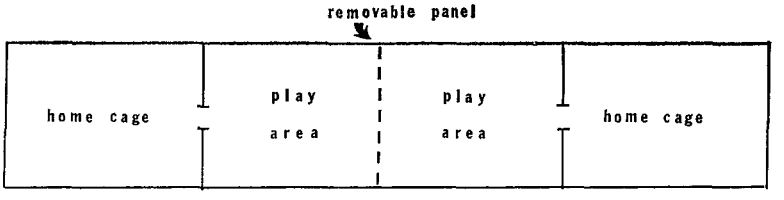


FIG. 1. Apparatus (top view).

each definition. In one instance, no correlation coefficient is given since the behavior occurred very infrequently during reliability sessions. All reliability coefficients are based on sessions conducted by the two observers during acquisition of developmental data for another project prior to initiation of the present study.

Infant–infant behaviors. The behaviors are defined as follows: (*a*) approach—oriented movement of 1 ft. or more toward another infant (.781); (*b*) withdraw—oriented movement of 1 ft. or more away from another infant (.825); (*c*) noncontact play—highly active, oriented movement toward or away from another infant involving two or more directional turns (also includes bee-line mock attack of three or more body lengths and rapid bouncing in place with orientation toward the other infant (.853); (*d*) contact play—wrestling or biting with rapid changes in location, or biting with head shaking, directed toward another infant (.954); (*e*) oral exploration—any mouthing, sucking, or licking on another infant (--); (*f*) ear-flip threat—flipping the ears with orientation toward another infant (.751); (*g*) jaw-drop threat—dropping the jaw with orientation toward another infant (.824); (*h*) clasp pull—clasping and jerking the hair or skin of another infant (.873); (*i*) bite—biting another infant (.887); (*j*) cuff—striking or hitting another infant (.899); (*k*) nonspecific contact—brushing against or touching another infant which is not included in any other score (.979); (*l*) cling—grasping or clutching the hair of another infant in such a way as to maintain contact (.987); (*m*) ventral contact—gross body contact with the ventral surface of another infant (.742); (*n*) gross body contact—gross contact with any surface of another infant's body except ventral (.954).

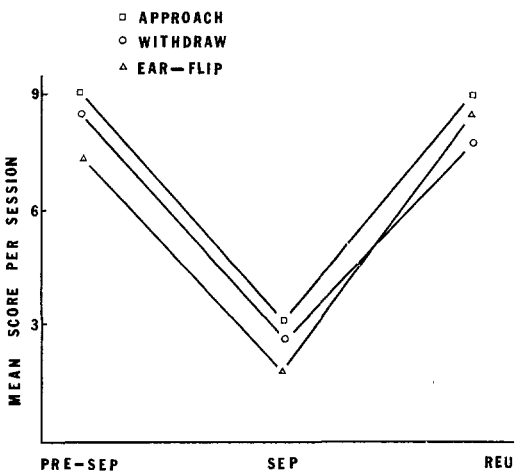


FIG. 2. Infant-infant approach, withdraw, and ear-flip threat for 3-wk.-separated infants. (Presep = preseparation or base-line period; sep = separation period; reu = reunion period.)

Infant nonsocial behaviors. The behaviors are defined as follows: (*a*) vocal screech—high-pitched screaming (.937); (*b*) vocal coo—tonal vocalization resulting in ooh-ooh sound (.894); (*c*) vocal yip—short, abrupt, high-pitched sound (.696); (*d*) self-mouthing—any mouthing or oral manipulation of infant's own body (.930); (*e*) oral exploration—mouthing or licking cage or other inanimate object (.894); (*f*) manual manipulation—manipulating or exploring the cage or other inanimate object with the hand (.774); (*g*) self-play—any behavior meeting the activity criterion of noncontact play, but without social orientation (.747).

Infant–mother behaviors. The following definitions describe infant–mother behaviors: (*a*) ventral contact—gross body contact with the ventral surface of the mother (.997); (*b*) cling—grasping or clutching the hair of the mother in such a way as to maintain contact (.992); (*c*) nipple contact—mouthing, sucking, or holding the nipple of the mother in the mouth (.979); (*d*) nonspecific contact—brushing against or touching the mother which is not included in any other score (.969); (*e*) approach—oriented movement of 1 ft. or more toward the mother (.784); (*f*) gross body contact—gross contact with any surface of the mother's body except ventral (.917).

Infant—other mother behaviors. The following describe infant–other mother behaviors: (*a*) nonspecific contact—as above, but directed or oriented toward adult female other than infant's own mother (.864); (*b*) withdraw—as above, but movement is away from adult female other than infant's own mother (.548).

Mother–infant behaviors. Mother–infant behaviors are defined as follows: (*a*) groom—spreading and picking at fur of infant (.973); (*b*) embrace—encircling or partly encircling the body of infant with one or both arms (.970).

Mother—other infant behaviors. The behaviors are defined as follows: (*a*) jaw-drop threat—as above, but directed by adult female toward infant other than her own (.938); (*b*) approach—as above, but movement is by adult female toward infant other than her own (.901).

Mother nonsocial behaviors. The behaviors are defined as follows: (*a*) lipsmack—smacking the lips rapidly with orientation toward the observer or without apparent orientation (.972); (*b*) jaw-drop threat—as above, but directed toward observer (.990).

Statistical Analysis

The mean score per session in each of the three experimental periods was calculated for each infant in each category of infant–infant and infant nonsocial behavior. Since mothers were not present for A groups during the separation period, the mean score per session for mother and mother-related behaviors was calculated for the preseparation and reunion periods. These data were then analyzed using repeated measures analysis

of variance and a priori comparisons were made using orthogonal coefficients. Wherever possible, each subject was used as its own control to minimize effects due to differences in the sexual composition of groups.

The behavior of and toward substitute mothers in SM groups was compared to the behavior of and toward real mothers in C groups during the separation period. The mean score per session for each category of mother and mother-related behaviors was calculated and analyzed by means of t tests. The .05 level of significance was used in all of the above tests.

Results

Effects of Separation

Analysis of preseparation and separation period differences revealed that the brief separation experienced by infants in the C groups produced virtually no change in behavior other than a significant increase in infant–infant contact play. In contrast, infants in the experimental groups (A and SM groups combined) showed very pronounced decreases during the separation period in infant playing and related behaviors. Statistically significant decreases were found in infant–infant approaches, withdraws, contact and noncontact plays, ear-flip and jaw-drop threats, clasp pulls, bites, cuffs, and nonspecific contacts (see Figure 2). The decrease in nonspecific contact was significantly greater for infants in the SM group than for infants in the A groups (Figure 3).

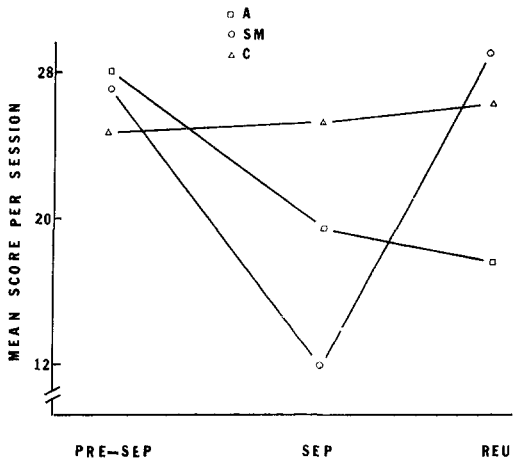


Fig. 3. Infant–infant nonspecific contact. (A = infants without adult female mother substitutes during separation period; SM = infants provided with adult female mother substitutes during separation period; C = control group composed of infants separated and immediately reunited with their own mothers during separation period; pre-sep = preseparation or base-line period; sep = separation period; reu = reunion period.)

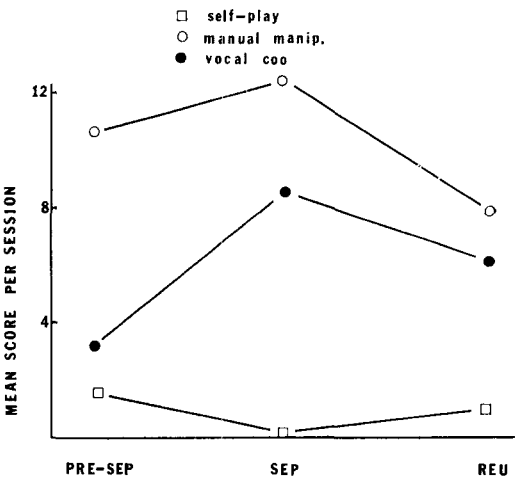


Fig. 4. Infant nonsocial self-play, manual manipulation, and vocal coo for 3-wk.-separated infants. (Pre-sep = preseparation or base-line period; sep = separation period; reu = reunion period.)

Infants in the experimental groups also showed significant changes during separation in some nonsocial behaviors as well. Self-playing decreased significantly while manual manipulation and vocal coo increased (Figure 4). Significantly different reactions to separation between the A and SM groups were found for self-mouthing and nonsocial oral exploration. Infants in the A groups showed a decrease in self-mouthing and an increase in nonsocial oral exploration during separation, while infants in the SM groups showed only slight changes in the behaviors (Figure 5). Vocal yip showed a significant decrease during separation only when the data over all groups was included.

Effects of Reunion

Few differences in infant–infant and infant nonsocial behavior were found during reunion between the groups as the behavior of infants in the experimental groups

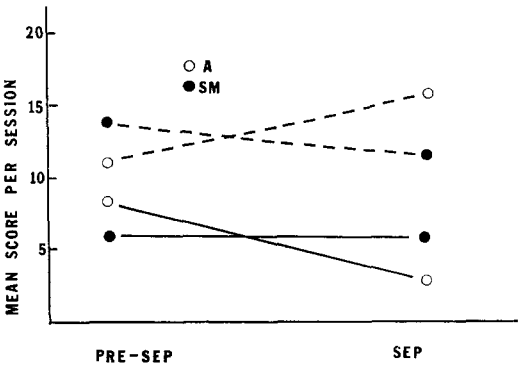


FIG. 5. Infant self-mouthing and nonsocial oral exploration. (A = infants without adult female mother substitutes during separation period; SM = infants provided with adult female mother substitutes during separation period; presep = preseparation or base-line period; sep = separation period.)

tended to return to preseparation levels (see Figure 2). There was a significant difference in reunion between the experimental and control groups in infant–infant clasp pull, a behavior which increased slightly in reunion but did not return to its preseparation level (Figure 6). There was also a significant difference in reunion between the A and SM groups for infant–infant nonspecific contact. This behavior returned to its preseparation level for the SM groups, but continued to decrease in reunion for the A groups (see Figure 3).

Analysis of reunion effects for mother and mother-related behaviors revealed an intensification of the mother–infant relationship for subjects in the experimental groups. Infant–mother ventral contact, cling, and nipple contact showed a significantly greater increase in reunion for the experimental than for the control groups. Also, infants in the A groups showed significantly greater increase in these behaviors than infants in the SM groups (Figure 7). The experimental groups combined also showed a significant differential change in mother–infant grooming during reunion. Compared to preseparation levels, mothers in the experimental groups groomed their infants more during reunion while mothers in the control groups groomed their infants less.

The return of mothers to reversed home cage locations resulted in a significant overall increase in infant–mother ventral contact, cling, and nipple contact. When only the data from the experimental groups were considered, the effect was significant only for infant–mother ventral contact and cling.

Behavior of and toward Substitute Mothers

Infants in the SM groups withdrew less from the adult female in the adjacent home cage during separation than did infants in the C groups whose real mothers were present in the separation period. Although no other significant differences were found, there was a tendency for real mothers to lipsmack more at the observer and to jaw-drop more at the other infant. There was also a nonsignificant tendency for infants to touch and make nipple contact more with real mothers. In fact, not a single instance of nipple contact with substitute mothers was observed.

DISCUSSION

Although a qualitatively similar reaction to maternal separation was seen in the rhesus macaque (Seay et al., 1962; Seay & Harlow, 1965; Spencer-Booth & Hinde, 1967) and in the pigtail (Kaufman & Rosenblum, 1967), some species-specific differences exist. In a recent study by Preston,

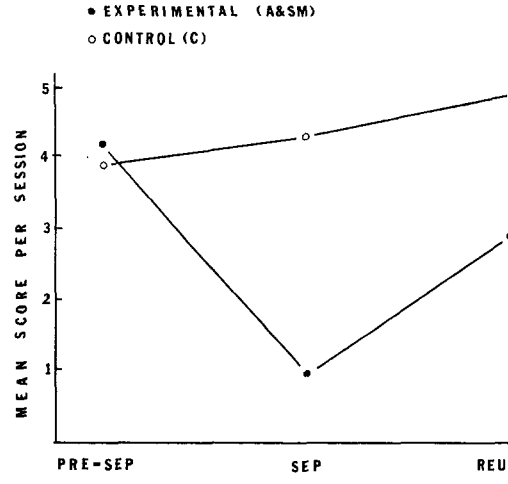


FIG. 6. Infant–infant clasp pull. (Pre-sep = preseparation or base-line period; sep = separation period; reu = reunion period.)

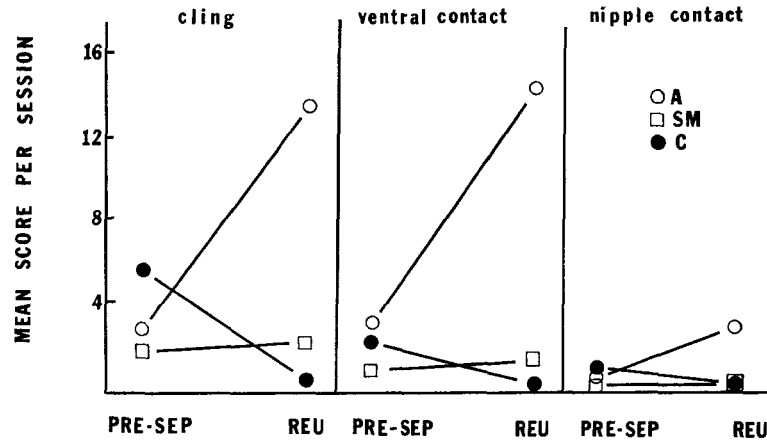


FIG. 7. Infant–mother cling, ventral contact, and nipple contact. (A = infants without adult female mother substitutes during separation period; SM = infants provided with adult female mother substitutes during separation period; C = control group composed of infants separated and immediately reunited with their own mothers during separation period; pre-sep = preseparation or base-line period; sep = separation period; reu = reunion period.)

Baker, and Seay (1970), 7-mo.-old patas (*Erythrocebus patas*) infants showed a slight nonsignificant increase in contact play during the separation period despite the usual decreases in noncontact and self-play, and a significant increase in contact play and close-quarters play behaviors during reunion. The investigators interpreted the difference between patas and macaque reactions to separation in terms of the relationship between maternal roles and peer interaction. Patas mothers are more protective than their Java counterparts and threaten the other infant more during close-quarters play. Separation, therefore, permitted behaviors which previously evoked threat reactions from the mother.

Species differences in the reaction to separation were also evident in the Rosenblum and Kaufman (1968) study. Bonnet macaques fail to show a severe reaction to separation if familiar substitutes are available since their social structure facilitates attachment behavior between infants and adult females other than the mother.

However, pilot work by these investigators indicated that either bonnet or pigtail infants, when placed in a group of unfamiliar adult conspecifics, barely survived a single day. Such brutality was not observed in the present study in which infants were housed with a single mother substitute. This apparent species difference may be due to the complex social situation employed by Rosenblum and Kaufman.

Comparisons with the rhesus data collected by Seay and colleagues in which a similar apparatus and procedure was employed indicate that although the reaction to separation was qualitatively similar the rhesus infants in those studies recovered more quickly than the Java infants in the present study. These data suggest that the mother–infant *attachment* is stronger in Java than in rhesus macaques despite the species similarity in *attachment behavior* observed by Seay (1967).

The reaction of the infants in the present study to separation was due primarily to loss of the specific mother to whom the infant had become attached. The effects observed could not be attributed to the seemingly traumatic nature of the separation procedure since infants who were separated and immediately reunited with mothers were minimally affected. In addition, the effects were considerable even when mother substitutes were provided. However, the better recovery during reunion shown by infants in SM groups suggests that an attachment between infants and mother substitutes was being formed. The data in-

dicate that Java infants are not monotropic, but will tend to establish relationships with other adults in the absence of the mother.

There was some indication that the effects of separation would have been less severe if mother substitutes had behaved in a more maternal manner. Although there were no statistically significant differences in the behavior of mothers and mother substitutes, substitutes appeared to be more indifferent toward infants. They seldom threatened the other infant and their attempts to punish infants housed with them, although infrequent, appeared to be more severe. However, one of the substitutes used in Group SM_2 frequently embraced the infant housed with her and appeared quite maternal. The infant involved appeared to recover somewhat from the effects of separation and several times attempted to play with the other infant in the group. These attempts resulted in the other infant screeching, fear grimacing, and withdrawing.

The present study also indicates that the attachment between a Java infant and its mother is not diminished by a 3-wk. separation period. In reunion, all infants were able to recognize and respond differentially to their own mothers after several minutes at most even when they were returned to reversed home cage locations. Infant playing and most related behaviors returned to preseparation levels shortly after the mother's return. But the increase in mother–infant and infant–mother positive behaviors, such as clinging and ventral contact, was evident throughout the 3-wk. reunion period, primarily for those infants who had not been provided with mother substitutes during separation. There was no indication of either detachment or aggression toward the mother as has sometimes been observed in human children following separation episodes (see Bowlby, 1952; Heinicke, 1956; Robertson, 1955).

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